Practical no 2 : Implementation of Control Statements

A) if, if-else, if-elif-else

Program 1: Check Whether Number is Positive

```
num = int(input("Enter a number: "))

if num > 0:
    print("Positive Number")
```

Output

```
Enter a number: 10
Positive Number
```

Program 2: Check Even or Odd (if-else)

```
num = int(input("Enter a number: "))

if num % 2 == 0:
    print("Even Number")
else:
    print("Odd Number")
```

Output

```
Enter a number: 7
Odd Number
```

Program 3: Find Greatest of Three Numbers (if-elif-else)

```
a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))
c = int(input("Enter Third Number: "))

if a > b and a > c:
    print("A is Greatest")
elif b > c:
    print("B is Greatest")
else:
    print("C is Greatest")
```

B) for and while Loop

Program 4: Print Numbers from 1 to 10 using for Loop

```
for i in range(1, 11):
    print(i)
```

Program 5: Display Multiplication Table using for Loop

```
num = int(input("Enter a Number: "))

for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

Program 6: Print Numbers from 1 to 10 using while Loop

```
i = 1

while i <= 10:
    print(i)
    i += 1
```

Program 7: Sum of First N Natural Numbers

```
n = int(input("Enter Value of N: "))

i = 1
sum = 0

while i <= n:
    sum += i
    i += 1

print("Sum =", sum)
```

C) break, continue, pass

Program 8: Demonstration of break

```
for i in range(1, 11):
    if i == 6:
        break
    print(i)
```

Output

```
1
2
3
4
5
```

Program 9: Demonstration of continue

```
for i in range(1, 11):
    if i == 5:
        continue
    print(i)
```

Output

```
1
2
3
4
6
7
8
9
10
```

Program 10: Demonstration of pass

```
for i in range(1, 6):
    if i == 3:
        pass
    print(i)
```

Output

```
1
2
3
4
5
```